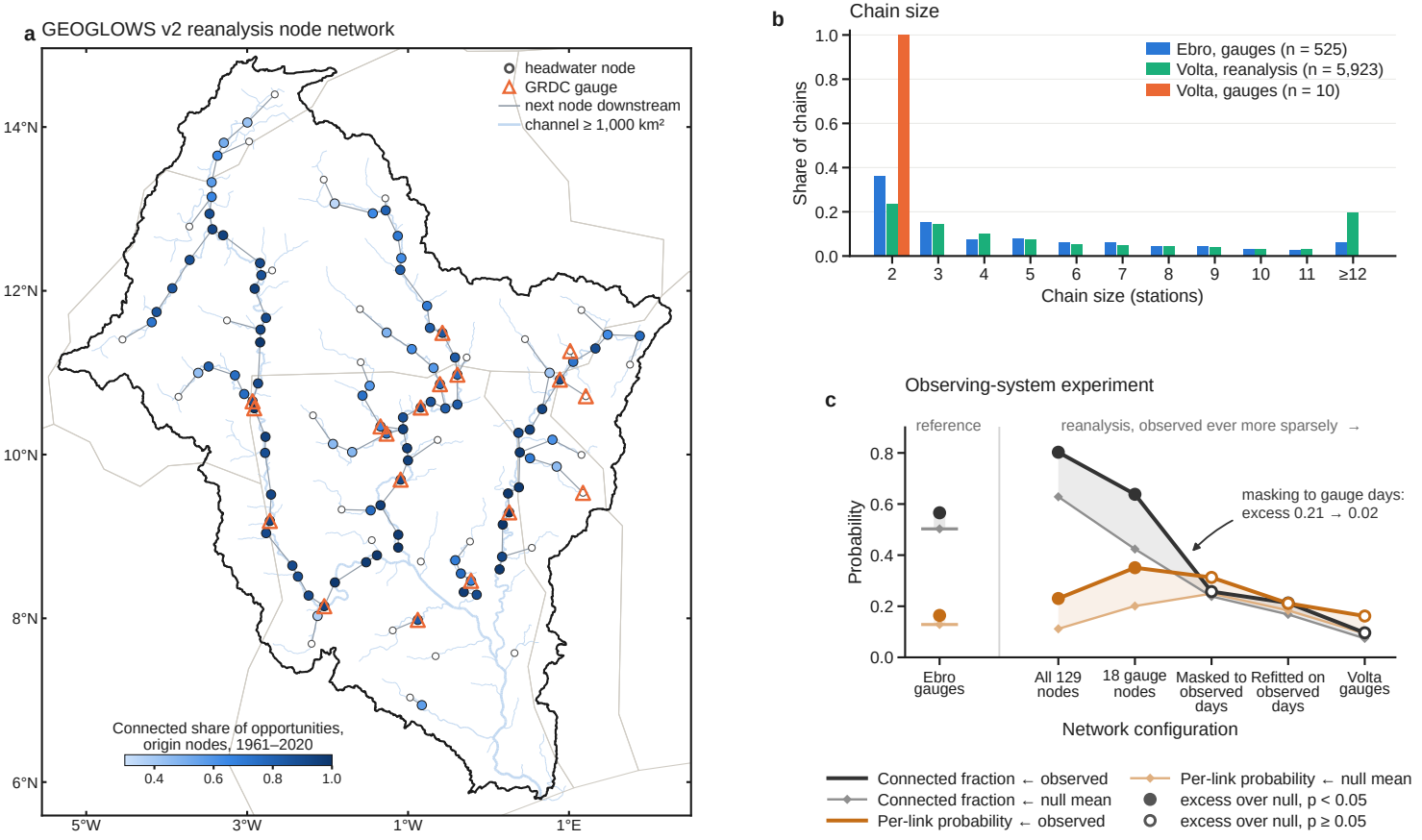

a GEOGLOWS v2 reanalysis node network
headwater node
GRDC gauge
next node downstream
channel ≥ 1,000 km²
14°N
12°N
10°N
8°N
6°N
5°W
3°W
1°W
1°E
Connected share of opportunities, origin nodes, 1961–2020
0.4
0.6
0.8
1.0
b
Chain size
Ebro, gauges (n = 525)
Volta, reanalysis (n = 5,923)
Volta, gauges (n = 10)
Share of chains
1.0
0.8
0.6
0.4
0.2
0.0
2
3
4
5
6
7
8
9
10
11
≥12
Chain size (stations)
c
Observing-system experiment
reference
reanalysis, observed ever more sparsely →
Probability
0.8
0.6
0.4
0.2
0.0
masking to gauge days:
excess 0.21 → 0.02
Ebro gauges
All 129 nodes
18 gauge nodes
Masked to observed days
Refitted on observed days
Volta gauges
Network configuration
Connected fraction ← observed
Connected fraction ← null mean
Per-link probability ← observed
Per-link probability ← null mean
excess over null, p < 0.05
excess over null, p ≥ 0.05